

Hip MRI — 30 Sample Cases

A worked teaching file of 30 common hip and pelvis MRI diagnoses with representative findings and impressions.

Labrum, FAI & Cartilage

Case 1 — Anterosuperior labral tear

Clinical: 34-year-old woman, recreational runner with 6 months of anterior groin pain and a positive FADIR (impingement) test.

F: On the radial and coronal oblique fat-suppressed PD sequences there is a linear high-signal cleft extending to the articular surface of the **anterosuperior labrum** at the **11–1 o'clock** position, consistent with a tear. The torn labral fragment measures **4 mm** in maximal dimension and there is mild blunting of the adjacent labral apex. Small **paralabral recess** without a discrete cyst. The chondrolabral junction is intact and the adjacent acetabular cartilage shows preserved thickness (**~2.5 mm**). Alpha angle on the oblique axial measures **48 degrees** (normal). No joint effusion beyond physiologic volume. Femoral head is spherical with normal marrow signal.

I: Articular-surface tear of the anterosuperior acetabular labrum (11–1 o'clock) without underlying cam or pincer morphology. No chondral loss.

Case 2 — Paralabral cyst

Clinical: 41-year-old man with deep aching lateral hip pain, worse with prolonged sitting.

F: A multiloculated **paralabral cyst** measuring **18 × 9 × 11 mm** arises from the **superior labrum (12–1 o'clock)** and dissects superolaterally along the acetabular rim, demonstrating T2-hyperintense fluid signal with thin internal septations and no internal enhancing solid tissue. The cyst communicates with an underlying **full-thickness labral tear** at its base. Mild adjacent acetabular rim edema. No nerve compression. Femoral head marrow normal; alpha angle **52 degrees**.

I: Superior paralabral cyst arising from an associated labral tear. The cyst confirms a clinically significant labral lesion.

Case 3 — Cam-type FAI with elevated alpha angle

Clinical: 26-year-old male collegiate hockey player with progressive anterior groin pain and reduced internal rotation.

F: There is an osseous bump at the **anterosuperior femoral head–neck junction** with loss of the normal concavity (pistol-grip morphology). The **alpha angle** measured on the oblique axial sequence is **68 degrees** (elevated; normal <55). A small **herniation pit** (3 mm subcortical cyst) is seen at the anterolateral head–neck junction. There is an associated articular-surface tear of the **anterosuperior labrum at 1 o'clock** and early **chondral delamination** of the adjacent acetabular cartilage. Femoral head–neck offset ratio is reduced. No significant pincer morphology.

I: Cam-type femoroacetabular impingement (alpha angle 68 degrees) with secondary anterosuperior labral tear and early acetabular chondral injury.

Case 4 — Pincer / mixed FAI

Clinical: 38-year-old woman, yoga instructor, with anterior and lateral hip pain at end-range flexion.

F: There is **acetabular overcoverage** with a lateral center-edge angle of **42 degrees** and a **crossover sign** indicating focal anterior overcoverage (cranial retroversion). The **alpha angle is mildly elevated at 58 degrees**, indicating a mixed cam-pincer pattern. The **anterosuperior labrum (12–2 o'clock)** is degenerated and intrasubstance torn with adjacent rim ossification. A **contrecoup** posteroinferior chondral signal abnormality is noted. No paralabral cyst.

I: Mixed (pincer-predominant) femoroacetabular impingement with labral degeneration/tear and early contrecoup chondral change.

Case 5 — Acetabular chondral delamination

Clinical: 31-year-old male soccer player with mechanical anterior hip pain and clicking.

F: MR arthrogram demonstrates contrast insinuating beneath the **anterosuperior acetabular cartilage** at the chondrolabral junction, producing a **delamination flap** over an arc of approximately **15 mm**. The overlying cartilage is detached but in situ with focal subchondral acetabular edema. Associated **anterosuperior labral tear at 1 o'clock**. Alpha angle **64 degrees** consistent with underlying cam morphology. Femoral cartilage preserved.

I: Anterosuperior acetabular chondral delamination with associated labral tear, in the setting of cam-type FAI.

Case 6 — Ligamentum teres tear

Clinical: 29-year-old woman after a fall with the hip in abduction; persistent deep groin pain and sense of instability.

F: The **ligamentum teres** is thickened and demonstrates intrasubstance and partial fiber discontinuity with surrounding fluid signal in the **fovea**, best seen on the coronal and axial fat-suppressed sequences, consistent with a **partial-thickness tear**. No complete avulsion from the fovea capitis. Mild synovitis. The acetabular labrum is intact. No fracture; femoral head marrow normal. Trace joint effusion.

I: Partial-thickness tear of the ligamentum teres. No labral tear or fracture.

Case 7 — Acetabular dysplasia with undercoverage

Clinical: 24-year-old woman with chronic anterolateral hip pain and a sense of instability, no trauma.

F: There is **acetabular dysplasia** with a shallow, undercovered acetabulum; the **lateral center-edge angle measures 18 degrees** (undercoverage; normal 25–40) and the acetabular index is increased. The **anterosuperior labrum is hypertrophied** (measuring up to 9 mm) and demonstrates an articular-surface tear at the 12–1 o'clock position from chronic overload. Mild superolateral acetabular subchondral marrow edema. No cam morphology (alpha angle 46 degrees). Femoral head congruent.

I: Developmental acetabular dysplasia with undercoverage (LCEA 18 degrees) and secondary hypertrophic labral tear from instability overload.

Case 8 — Gluteus medius/minimus tendinosis

Clinical: 57-year-old woman with chronic lateral hip pain and tenderness over the greater trochanter.

F: The **gluteus medius and minimus tendons** demonstrate intermediate-to-increased intrasubstance signal and mild thickening at their insertions on the **lateral and superoposterior facets of the greater trochanter**, without a discrete fluid-signal gap to indicate a tear. Mild peritrochanteric soft tissue edema. No tendon retraction and no fatty muscle atrophy. Trochanteric bursa shows only trace fluid.

I: Gluteus medius and minimus insertional tendinosis without tear. Findings of greater trochanteric pain syndrome.

Case 9 — Gluteal tendon tear with fatty atrophy

Clinical: 68-year-old woman with a chronic limp, Trendelenburg gait, and refractory lateral hip pain.

F: There is a **full-thickness tear of the gluteus medius tendon** at the lateral facet insertion with a fluid-filled gap measuring **12 mm** and **8 mm of tendon retraction**, best demonstrated on coronal fat-suppressed T2. The gluteus minimus tendon is partially torn. Moderate **fatty atrophy (Goutallier grade 2–3)** of the gluteus medius and minimus muscle bellies. Distended trochanteric bursa. No femoral neck fracture.

I: Full-thickness gluteus medius tendon tear with retraction and moderate fatty muscle atrophy ("rotator cuff of the hip" tear); partial gluteus minimus tear.

Case 10 — Greater trochanteric bursitis / GTPS

Clinical: 52-year-old woman, runner, with point tenderness lateral to the greater trochanter and pain lying on that side.

F: A distended **greater trochanteric (subgluteus maximus) bursa** contains T2-hyperintense fluid extending superficial to the greater trochanter over a craniocaudal length of **5 cm**, with surrounding peritrochanteric soft tissue edema. The underlying **gluteus medius and minimus tendons are intact** with only mild tendinosis. No tendon tear, retraction, or fatty atrophy. No osseous lesion of the trochanter.

I: Greater trochanteric bursitis with mild abductor tendinosis — greater trochanteric pain syndrome (GTPS). No tendon tear.

Case 11 — Proximal hamstring origin tear/avulsion

Clinical: 45-year-old man, water-skiing, acute posterior thigh pain with a "pop" while the hip flexed and knee extended.

F: At the **ischial tuberosity** there is a tear of the **conjoint (semitendinosus–biceps femoris) hamstring origin** with a fluid-filled gap and **3 cm of tendon retraction**; the semimembranosus origin remains partially attached. A large surrounding hematoma tracks distally along the posterior thigh. The **sciatic nerve** lies immediately adjacent to the retracted stump without entrapment at present. No avulsed osseous fragment.

I: High-grade (near-complete) proximal hamstring tendon avulsion from the ischial tuberosity with 3 cm retraction. Sciatic nerve adjacent but not entrapped.

Case 12 — Rectus femoris strain

Clinical: 22-year-old male sprinter with sudden anterior thigh/groin pain during acceleration.

F: There is a **grade 2 strain of the rectus femoris** at the proximal myotendinous junction involving the central tendon, with feathery intramuscular T2-hyperintense edema and a small amount of fluid along the central aponeurosis; estimated cross-sectional involvement **~35%**. No complete tendon discontinuity and no avulsion at the anterior inferior iliac spine (AIIS). Adjacent small intramuscular hematoma. Sartorius and iliopsoas intact.

I: Grade 2 rectus femoris strain at the proximal myotendinous junction without full-thickness tear or AIIS avulsion.

Case 13 — Iliopsoas tendinopathy / bursitis

Clinical: 36-year-old woman, dancer, with anterior groin pain and a painful snapping sensation with hip extension.

F: The **iliopsoas tendon** is mildly thickened with increased intrasubstance signal at the level of the **lesser trochanter**, and there is a distended **iliopsoas bursa** containing T2-hyperintense fluid measuring **14 mm** anterior to the hip capsule, communicating with the joint. No tendon tear. Mild reactive marrow signal at the lesser trochanter insertion. No labral tear identified on these non-arthrographic images.

I: Iliopsoas tendinopathy with iliopsoas bursitis, compatible with internal snapping hip syndrome.

Case 14 — Adductor strain

Clinical: 28-year-old male soccer player with acute medial groin pain after a lunging tackle.

F: There is a **grade 2 strain of the adductor longus** at its origin from the pubic body, with myotendinous edema, feathery T2 hyperintensity, and a small interstitial fluid cleft; no full-thickness avulsion from the pubis. Mild edema also involves the adductor brevis. The **rectus abdominis–adductor aponeurosis** is intact and there is no secondary cleft sign. No osseous avulsion. Symphysis pubis unremarkable.

I: Grade 2 adductor longus strain at the pubic origin without complete avulsion. No associated athletic pubalgia findings.

Bone & Marrow

Case 15 — Femoral head AVN with double-line sign

Clinical: 44-year-old man on chronic corticosteroids with insidious groin pain.

F: A serpiginous subchondral lesion in the **anterosuperior femoral head** demonstrates the **"double-line sign"** on T2 (inner hyperintense granulation tissue, outer hypointense sclerotic rim) and a corresponding low-signal rim on T1, involving an estimated **30% of the weight-bearing surface**. The femoral head remains spherical without subchondral collapse or crescent sign (**ARCO stage II**). A similar but smaller lesion is present in the contralateral head on the limited views. No joint effusion of note.

I: Avascular necrosis of the right femoral head, ARCO stage II (no collapse), with characteristic double-line sign; likely steroid-related and bilateral.

Case 16 — Femoral neck stress fracture

Clinical: 19-year-old female military recruit with progressive groin pain on weight-bearing after increased marching.

F: Along the **medial (compression-side) cortex of the femoral neck** there is a hypointense fracture line on T1 surrounded by extensive marrow edema on fat-suppressed T2, extending over approximately **40% of the neck width**. No fracture line extends to the lateral (tension) cortex and there is no displacement. Periosteal edema present. Femoral head marrow otherwise normal; no AVN.

I: Compression-side femoral neck stress fracture (<50% width) without tension-side extension or displacement. Lower-risk pattern, but protected weight-bearing warranted.

Case 17 — Subchondral insufficiency fracture

Clinical: 71-year-old woman with osteopenia and acute onset of severe hip pain without trauma.

F: A **subchondral insufficiency fracture** is seen as a low-signal line paralleling the subchondral bone plate of the **superior femoral head**, with overlying marrow edema and mild subchondral irregularity; the low-signal line measures **16 mm** in length. There is mild associated femoral head flattening (<2 mm). No serpiginous double-line sign to suggest AVN. Joint effusion present. Acetabular cartilage preserved.

I: Subchondral insufficiency fracture of the superior femoral head with associated marrow edema and early subchondral flattening. Distinguished from AVN by the absence of a double-line sign.

Case 18 — Transient osteoporosis / bone marrow edema syndrome

Clinical: 39-year-old man with self-limited onset of hip pain and antalgic gait.

F: Diffuse **bone marrow edema** involves the **femoral head and neck**, extending into the intertrochanteric region, low signal on T1 and high signal on fat-suppressed T2, without a subchondral fracture line, double-line sign, or focal collapse. A reactive joint effusion is present. The acetabulum and surrounding soft tissues are normal. No discrete lesion.

I: Transient osteoporosis of the hip (bone marrow edema syndrome). Absence of a fracture line or double-line sign argues against insufficiency fracture and AVN; typically self-limited.

Case 19 — Hip osteoarthritis

Clinical: 64-year-old man with chronic groin pain, morning stiffness, and reduced range of motion.

F: There is **superolateral joint-space narrowing** with full-thickness chondral loss, subchondral cysts in the acetabular roof (largest **8 mm**) and femoral head, subchondral sclerosis, and marginal **osteophytes** at the femoral head-neck junction and acetabular rim. The labrum is degenerated with intrasubstance signal and small tears. Mild reactive bone marrow edema flanks the weight-bearing surfaces. Moderate joint effusion.

I: Moderate-to-advanced osteoarthritis of the hip with superolateral cartilage loss, subchondral cysts, osteophytosis, and degenerative labral change.

Case 20 — Occult / intertrochanteric fracture

Clinical: 78-year-old woman after a low-energy fall, unable to bear weight; radiographs negative.

F: A nondisplaced **intertrochanteric fracture line** extends obliquely across the proximal femur, hypointense on T1 with surrounding marrow edema and high signal on fat-suppressed T2 (STIR), extending from the greater toward the lesser trochanter without cortical displacement. A small adjacent soft tissue hematoma is present. No extension into the femoral neck or head; no AVN-type signal.

I: Radiographically occult nondisplaced intertrochanteric femoral fracture. MRI confirms acute fracture not seen on radiographs.

Pelvis & Other

Case 21 — Athletic pubalgia / sports hernia with secondary cleft

Clinical: 27-year-old male professional soccer player with chronic lower abdominal/groin pain aggravated by kicking and sit-ups.

F: There is disruption of the **rectus abdominis–adductor longus aponeurosis** at its attachment to the anterior pubis with a **secondary cleft sign** — fluid signal tracking inferolaterally from the symphysis along the adductor origin — on the coronal and axial fat-suppressed images. Associated **adductor longus origin tendinosis** and mild pubic marrow edema. No true inguinal hernia sac. Symphyseal alignment maintained.

I: Athletic pubalgia ("sports hernia") with rectus abdominis–adductor aponeurotic injury and secondary cleft sign. No fascial inguinal hernia.

Case 22 — Osteitis pubis

Clinical: 33-year-old male distance runner with chronic central groin pain and symphyseal tenderness.

F: Symmetric **subchondral bone marrow edema** flanks both sides of the **symphysis pubis** on fat-suppressed T2, with subchondral irregularity, marginal sclerosis, and a widened/degenerated symphyseal fibrocartilage. Small subchondral cysts are present bilaterally. No discrete adductor avulsion and no secondary cleft. No drainable collection or aggressive features.

I: Osteitis pubis with bilateral symphyseal subchondral edema and degenerative change, consistent with chronic overuse.

Case 23 — Sacroiliitis

Clinical: 25-year-old man with inflammatory-pattern buttock pain, morning stiffness, and elevated CRP; query axial spondyloarthropathy.

F: There is **periarticular bone marrow edema** on STIR involving the **iliac and sacral sides of both sacroiliac joints**, more on the iliac aspect, with subchondral high signal abutting the joint. Early erosions and subchondral sclerosis are present, and small foci of **fatty metaplasia** on T1 indicate chronicity. No ankylosis yet. Findings are bilateral and roughly symmetric.

I: Active bilateral sacroiliitis with chronic structural changes, compatible with axial spondyloarthropathy in the appropriate clinical setting.

Case 24 — Sacral insufficiency fracture

Clinical: 74-year-old woman with osteoporosis and low back/buttock pain without trauma.

F: Vertically oriented bands of **bone marrow edema** parallel the **sacral ala bilaterally** with hypointense fracture lines on T1, and a transverse component crosses the sacral body, producing the classic **"H-pattern" (Honda sign)**. No soft tissue mass or aggressive destruction. The SI joints are unremarkable. No retropulsion or neural foraminal compromise.

I: Bilateral sacral alar insufficiency fractures with H-shaped (Honda) pattern, characteristic of insufficiency fracture in osteoporotic bone.

Case 25 — Deep gluteal / piriformis syndrome with sciatic nerve

Clinical: 48-year-old woman with deep buttock pain radiating down the posterior thigh, worse with sitting.

F: The **piriformis muscle** is asymmetrically enlarged with mild intramuscular edema, and the **sciatic nerve** in the deep gluteal space demonstrates increased T2 signal and mild caliber change as it courses adjacent to the inferior border of the piriformis. No mass lesion or hematoma compressing the nerve. Ischiofemoral interval preserved; quadratus femoris normal in signal.

I: Deep gluteal (piriformis) syndrome with sciatic nerve irritation. No compressive mass; correlate clinically.

Case 26 — Muscle strain / contusion

Clinical: 30-year-old man after a direct blow to the lateral thigh during contact sport ("dead leg").

F: There is a **contusion of the vastus lateralis** with feathery intramuscular T2-hyperintense edema and a small interstitial hematoma, without a discrete full-thickness tear or large fluid collection; involvement is estimated at **20%** of the muscle cross-section. The overlying fascia is intact and there is no myotendinous junction gap. No osseous injury of the femur. No findings of myositis ossificans at this stage.

I: Vastus lateralis contusion with intramuscular edema and small hematoma; no significant muscle tear. Follow-up if calcification/myositis ossificans is suspected later.

Case 27 — Labral degeneration in osteoarthritis

Clinical: 61-year-old woman with chronic hip pain and known early OA.

F: The **acetabular labrum** is diffusely thickened with globular intrasubstance T2 signal that does not clearly reach the articular surface, along with **labral degeneration** and small intrasubstance tears at the anterosuperior and superior margins. There is accompanying superolateral chondral thinning and marginal osteophytes. A small **paralabral ganglion (6 mm)** is present. No acute displaced labral fragment.

I: Degenerative labral disease in the setting of hip osteoarthritis, with intrasubstance signal and small tears rather than a discrete acute tear.

Case 28 — Ischiofemoral impingement

Clinical: 55-year-old woman with chronic deep buttock/groin pain, worse with extension and adduction.

F: The **ischiofemoral interval** (between the lateral ischium and lesser trochanter) is narrowed to **12 mm** and the **quadratus femoris space** to **8 mm**, with **edema and partial-thickness tearing of the quadratus femoris** muscle on fat-suppressed T2. No fatty atrophy of the quadratus femoris yet. Hamstring origin intact. No mass.

I: Ischiofemoral impingement with narrowed ischiofemoral and quadratus femoris spaces and quadratus femoris edema.

Case 29 — Apophyseal avulsion (AIIIS)

Clinical: 16-year-old male sprinter with acute groin pain and a "pop" while accelerating from the blocks.

F: There is a displaced **avulsion of the anterior inferior iliac spine (AIIS) apophysis** at the **direct head of the rectus femoris origin**, with a bony fragment displaced approximately **9 mm** and surrounding marrow and soft tissue edema. The unfused apophysis is widened relative to the contralateral side. Adjacent rectus femoris tendon intact at its bony attachment. No intra-articular extension.

I: Acute apophyseal avulsion of the AIIS at the rectus femoris direct-head origin, displaced 9 mm — a typical adolescent athletic injury at the open apophysis.

Case 30 — Hip effusion with proliferative synovitis

Clinical: 50-year-old man with progressive hip pain, stiffness, and a sensation of fullness; query inflammatory synovial process.

F: There is a moderate **hip joint effusion** with diffuse **synovial thickening**, including frond-like low-signal foci on gradient images that may reflect hemosiderin or synovial proliferation. No discrete calcified intra-articular loose body and no aggressive osseous erosion beyond mild pressure remodeling of the femoral neck. Labrum and cartilage are otherwise preserved. No fracture or marrow edema.

I: Hip joint effusion with proliferative synovitis; differential includes pigmented villonodular synovitis (PVNS)/synovial chondromatosis versus inflammatory arthropathy — correlate and consider synovial biopsy.